

STRATEGIC PLANING FOR DATA DRIVEN LOGISTICS OPTIMIZATION

PROJECT 1 – LOGISTICS DATA ANALYSIS

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DATASET USED – DELIVERY LOGISTICS DATA

RECORDS ANALYSED – 25,000

PRIMARY BUSINESS PROBLEM – REDUCE DELIVERY DELAYS, WHILE
CONTROLLING LOGISTICS COST AND IMPROVED CUSTOMER
SATISFACTION.

TOOLS USED – Python, Pandas, Pycharm, Matplotlib

INTRODUCTION

There are a lot of different aspects regarding logistics and last-mile delivery activities that can impact distribution reliability, costs, and satisfaction levels of customers. Weather, distance of transportation, type of delivery, characteristics of the vehicles used, and other aspects are crucial for the achievement of a company's logistic objective.

Data Science enables analysis of historical data in order to identify tendencies related to operating problems and support research-driven decisions. Logistics Performance Index of the World Bank states the importance of logistics service quality and costs competitiveness as well as something called tracing and tracking and the share of deliveries coming on time.

The project has the goal of creating a strategic data science method of logistics performance analysis. In order to solve this issue, a dataset of 25,000 delivery operations will be examined, including delivery delays, costs, distance, weather, delivery types, and customer ratings.

The main objective of the project is to achieve an understanding of logistics performance; find out the causes of delivery delays, and create a plan of further data preparation analysis including data cleansing, explorative analysis, predictive analytics, and optimization of logistics processes.

PROBLEM DEFINITION

The main problem addressed in this work is related to delivery delays occurring in logistics. Delays can have an impact on efficiency of processes, delivery costs and customer satisfaction.

The available dataset contains information about the delivery process, conditions of transportation, type of delivery, weather conditions, distance, cost of delivery, customer ratings and delivery result. The mentioned indicators will be used by the project to reveal patterns indicating when delivery occurs late and clarify what factors should be taken into account in the future research.

The key analytical question is the following:

- What operational and environmental factors are relevant to delays in delivery and how can the information received from data be used for the improvement of logistics performance?
- After this, the project is going to apply this analysis to predictive modelling when historical data on delivery is analysed so as to calculate the probability of the delivery taking place late.

PROBLEM OBJECTIVES

This project features the following goals:

1. Set a reference point for logistics performance, including delivery delay proportion, failures proportion, delivery expenses, distance travelled, and ratings by customers.
2. Find out the reasons behind delays, which could be linked to weather conditions, means of delivery, types of vehicles, areas of destination, delivery partners, and distance travelled.
3. Establish the connection between delays and results of business practises, relating them especially with delivery expenses and customer orientation scores.
4. Create plans for employing various tools of data analysis for predicting delays.
5. Construct solutions for minimizing delays and improving customer service along with managing delivery costs.

BASELINE KPI'S

KPI	Baseline	Purpose
Total Deliveries	25,000	Measures dataset delivery volume
Delivery Delay Rate	26.68%	Measures proportion of delayed deliveries
Delivery Failure Rate	5.31%	Measures unsuccessful delivery proportion
Average Delivery Cost	864.94	Measures average logistics cost
Average Distance	150.39 km	Measures transportation distance
Average Customer Rating	3.67/5	Measures customer satisfaction

Table1. KPI Table

The results of the baseline analysis show that 26.68% of deliveries are delayed and 5.31% are failed. The average cost of delivery was 864.94 and the average distance was 150.39 km. As a result, the average rating by customers were 3.67 out of 5.

These indicators give us a point of reference for measuring success in future implementations of the strategies. The same indicators can be used later in the project to measure success in improving the processes of delivery reliability, cost-efficiency, and satisfaction level of customers.

DATASET DESCRIPTION

This project includes a logistics delivery data set that has 25,000 delivery records. The delivery data set is a collection of records for an observation of delivery with details about the delivery partner, packaging, vehicle, mode of delivery, area, weather, delivery distance, delivery timings, delivery outcomes, customer review and delivery costs.

The data set is made up of categorical and numerical variables. The categorical variables include delivery partner, type of package, type of vehicle, mode of delivery, area of delivery, and weather conditions as well as delivery status. The numerical variables include delivery distance, package weight, delivery rating and delivery cost. The delay variable acts as a binary target variable to show if the delivery is late or not.

This filled dataset gives enough information to perform preliminary strategy analysis for the delivery results and further to be used for exploratory analysis and prediction modelling.

Variable	Type	Analytical Role
delivery_id	Numeric	Delivery identifier
delivery_partner	Categorical	Operational factor
package_type	Categorical	Package characteristic
vehicle_type	Categorical	Transportation factor
delivery_mode	Categorical	Service-level factor
region	Categorical	Geographic factor
weather_condition	Categorical	Environmental factor
distance_km	Numeric	Transportation distance
package_weight_kg	Numeric	Package characteristic
delivery_time_hours	String	Delivery timing
expected_time_hours	String	Expected delivery timing
delayed	Boolean	Primary target variable
delivery_status	Categorical	Delivery outcome
delivery_rating	Numeric	Customer satisfaction
delivery_cost	Numeric	Cost/efficiency measure

Table 2. Columns Data Types

DATA QUALITY ASSESSMENT

Before convincing the more thorough analysis of the data quality, a preliminary assessment was done. The review comprised data types, number of missing data, and also, duplicate identifiers as well as variable and variable correlation with the target variable.

6.1 Data Types

It was determined that some variables needed to be preprocessed again. However, the `delivery_time_hours` and `expected_time_hours` were imported as character variables rather than numerical ones. During the investigation process, it was revealed that they both had the same value (00:00.0) throughout the dataset. Thus, it can be concluded that these variables do not have any useful variation for the analysis.

6.2 Delivery-Time Limitation

The delivery-time variables are not regarded as missing values by Pandas but still, they do not carry any analytical value because they remain constantly equal to each other (00:00.0). If these values are converted in numbers (zeros), the data on the delivery time would not be considered useful and might even mislead the analysis. Therefore, these variables will not be used in this research and the limitation of the dataset will be mentioned.

6.3 Duplicated Delivery IDs

In this case, the size of our dataset amounts to 25000 records with 24502 unique delivery ids. However, there are ids repeated 250 times while the total number of duplicates reached 498 since no row has exactly the same value twice in the record. So, in this case, the duplicates, as records, shouldn't be discarded because they still may correspond to different points associated with the same id. This will be implemented in the data cleaning and preprocessing stages of the research.

Data-quality check	Result
Total records	25,000
Unique delivery IDs	24,502
Repeated ID occurrences	498
Exact duplicate rows	0

Table 3. Data Quality Check

6.4 Leakage of the Target Variable

The relationship between `delivery_status` and `delayed` was found to be straight-forward as the status “Delivered” corresponds to False in terms of `delayed`, while any record with “Delayed” or “Failed” statuses have True in terms of `delayed` variable which means that `delivery_status` contains information about the target itself. Thus, this feature must not be used in the process of model building for predicting delays because its presence will mean that the model would automatically get access to information about the dependent variable, thus revealing the leakage of the target.

Issue	Finding	Planned Action
Time variables	Constant 00:00.0 values	Exclude from current analysis; document limitation
Delivery IDs	Two IDs repeated 250 times each	Investigate during preprocessing
Exact duplicates	None identified	No automatic row removal
<code>delivery_status</code>	Directly associated with target	Exclude from delay-prediction features
Data types	Time fields stored as strings	Review during preprocessing

Table 4. Data Leakage Table

In summary, the chosen dataset fits well for the first strategic evaluation since it has as many as 25,000 observations, and there are lots of operational, environmental, and financial parameters. At the same time, the discovered problems with data quality must be kept in mind before making predictions.

INITIAL DATA ANALYSIS AND BUSINESS INSIGHTS

7. Preliminary Data Analysis and Business Knowledge

The initial exploratory analysis provided insights into the occurrences of delivery delays and their potential interdependence with the conditions of operations and environment. The examination was focused on delivery delay percentages determined by such weather conditions or delivery methods as well as the distinctions in transport distance, cost, and satisfaction level between delayed and on-time deliveries.

7.1 General Delivery Efficiency

As per the conducted analysis of the 25,000 deliveries studied, the amount of delayed deliveries was revealed as 6,669. The total percentage of all the delays was found to be 26.68%. Moreover, another 1,328 deliveries were defined as failed, leading to the failure percentage of 5.31%. The average delivery cost amounts 864.94, while the average delivery distance is defined as 150.39 km. The satisfaction level according to customers' opinions is rated as 3.67 out of 5 as an average result.

KPI	Value
Total Deliveries	25000.00
Delayed Deliveries	6669.00
Delay Rate (%)	26.68
Failure Rate (%)	5.31
Average Delivery Cost	864.94
Average Distance (km)	150.39
Average Customer Rating	3.67

Table 5. KPI Table

7.2 Weather Conditions and Delivery Delays

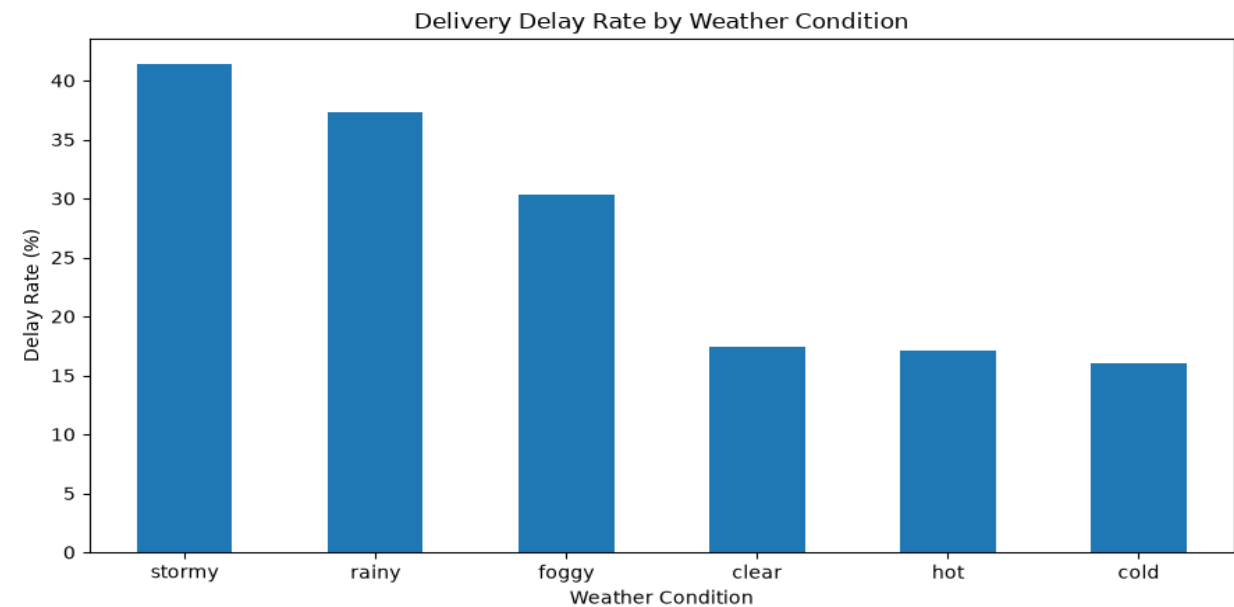


Figure 1. Delivery Rate by Weather Condition

The weather conditions have an important influence on delivery delays. Stormy weather led to the highest delay amount with a rate of 41.45%. Rainy weather conditions are next with a delay percentage of 37.35% followed by foggy conditions at 30.32%. Clear, hot and cold weather condition led to much lower delays with scores of 17.43%, 17.12%, and 16.02% respectively.

The findings imply that adverse weather conditions must be included in the planning of delivery operations. Weather-aware route planning, contingency resources, and proactive customer communication may enable logistics companies to operate under higher levels of risk.

7.3 Delivery Mode and Delays

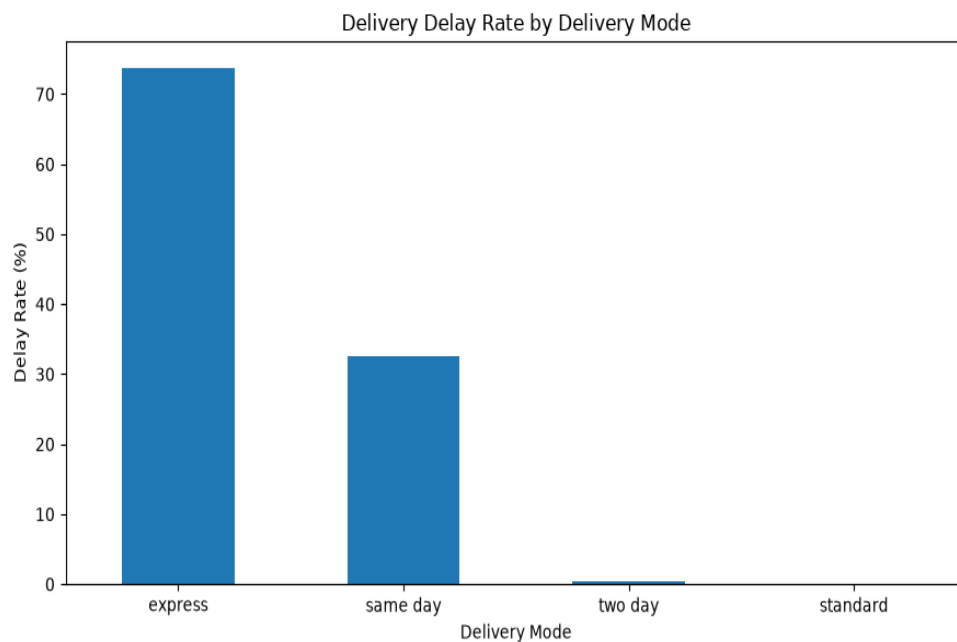


Figure 2. Delay Rate by Delivery Mode

Delivery mode is one of the delivery types with one of the largest variances in delay rates observed. Express deliveries have an observed delay rate of 73.78%, while same-day delivery has a delay rate of 32.54%. Two-day deliveries have a significantly lesser delay rate of 0.43%, while standard deliveries have no delays recorded in the dataset.

The unusually large discrepancy in delay rates across delivery types is concerning. One potential cause of the explained discrepancy can be that express and same-day delivery types are characterized by higher delivery-time limitations. However, the analysis does not make any conclusions about causality. Further research should focus on whether other factors, such as weather, distance, and operational specifics, play a role in the discrepancy.

7.4 Distance and Delivery Delay

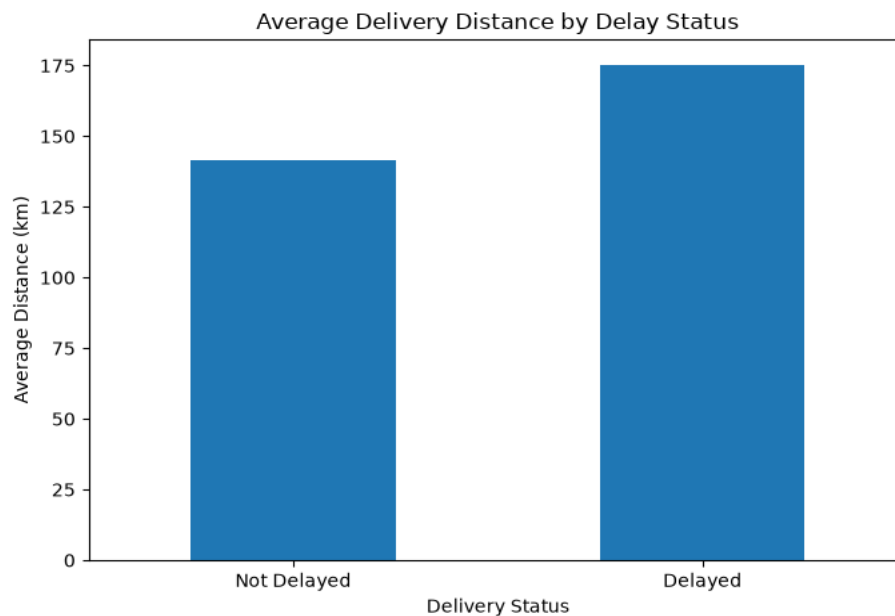


Figure 3. Average Delivery Distance by Delay

The analysis showed that the delivery distance varies considering the delay in delivery. Average distance travelled in case of delayed deliveries is 175.37 km as opposed to 141.30 km for non-delayed deliveries. This means that delivery distance in case of delayed delivery is around 24% higher.

Long delivery distances can thus be considered as an important risk factor causing delays in delivery, and the issue can be analysed in further detail by means of correlation analysis and predictive modelling later in the project.

7.5 Delivery Cost and Customer Satisfaction

Metric	Not Delayed	Delayed
Average Delivery Cost	809.75	1,016.65
Average Distance (km)	141.30	175.37
Average Customer Rating	4.21	2.18

Table 6. Cost and Customer Satisfaction

On the comparison of both delayed deliveries and non-delayed deliveries, some differences can be observed when it comes to the cost of delivery as well as its customer satisfaction. It has been realized from the results of delivery comparison that average delivery cost of \$1,016.65 for the delayed deliveries while non-delayed deliveries cost only \$809.75. Likewise, another statistic that has been found out as the average rating of deliveries is greatly affected by delays as well. More specifically, it has been found out that average rating of the deliveries which were delayed is only 2.18 which is a little bit lower than average rating of deliveries without delays which equals to 4.21.

These results show the relationship between the delivery delays and delivery costs along with such an important factor as customer ratings. Therefore, from the perspective of the business, lowering delivery time can lead to the main advantages of the company apart from efficiency in terms of operations.

7.6 OTHER OPERATIONAL FACTORS

The first examination of the types of vehicles and the regions shows that very little difference has occurred in terms of the delivery delays. The vehicles have had delay rates within the limits of approximately 26.14% to 27.04%, and the regions have had delay rates within the limits of approximately 25.80% to 27.25%. The average package weight has also remained almost unchanged when comparing the delayed and non-delayed deliveries, being equal to approximately 25.17 kgs and 25.14 kgs for the referred types of deliveries, correspondingly.

Considering the previously discussed factors (i.e., weather, delivery mode and transport distance), it can be said that these factors are not as significant in terms of the delivery delay issue. Nevertheless, the relevance of these measurements should be justified in a more specialized analysis with the help of multivariate analysis rather than rejected based on the previous comparisons made.

7.7 SUMMARY OF INITIAL FINDINGS

Factor	Initial Finding	Priority
Weather	Stormy and rainy conditions have substantially higher delay rates	● High
Delivery Mode	Express and same-day have much higher delay rates	● High
Distance	Delayed deliveries average ~24% greater distance	● High
Delivery Cost	Delayed deliveries have higher average cost	● Medium–High
Customer Rating	Delayed deliveries have substantially lower ratings	● Medium–High
Vehicle Type	Very small differences in delay rates	● Lower
Region	Relatively small differences	● Lower
Package Weight	Almost no difference	● Lower

8. Strategic Analytical Roadmap

The project aims to adopt a structured data-science lifecycle moving from strategic definition of the problem to the final phases of analysing the data, exploring and making predictions, and optimizing the operational processes. Each step relies on the results of the previous one.

8.1 Overall Data science workflow

Phase	Project	Main Activities	Expected Output
1. Strategic Planning	Project 1	Define business problem, objectives, KPIs, research questions and analytical strategy	Strategic project plan
2. Data Cleaning & Preparation	Project 2	Handle data-quality issues, validate variables, investigate duplicates and prepare features	Clean analytical dataset
3. Exploratory Data Analysis	Project 3	Analyze distributions, relationships, patterns and visualize key findings	EDA report and visualizations
4. Predictive Modelling	Project 4	Build and evaluate machine-learning models for delay prediction	Predictive model
5. Optimization & Recommendations	Project 4	Translate model insights into operational strategies	Logistics optimization recommendations

8.2 Project-2. Project Data Cleaning and Preparation

The second phase is going to emphasize the preparation of the data required for reliable analysis and modelling. Findings regarding data quality issues will be verified in a systematic manner, instead of making assumptions about the data.

Future Actions:

- Make sure column types are correct
- Investigate fields with constant time
- Inspect repeated delivery IDs
- All missing values need to be found
- Look for outliers
- Test categorical consistency
- Check distributions
- Use encoding on categorical data when needed
- Build final dataset for modelling purposes.

Important law:

Removing records only because there are repeated delivery IDs is not allowed, any possible removal of records from the dataset or other changes will have a solid reason based on thorough investigation of the data quality issues.

8.3 Project – 3. Exploratory Data Analysis

During the third stage, a detailed exploratory analysis is performed to find out how delivery delays are impacted by various operational factors.

In particular analysis will cover:

- Weather and delay
- Mode of delivery and delay
- Distance and delay
- Type of vehicle and delay
- Region and delay
- Delivery partner and delay
- Characteristics of package and delay
- Delay and delivery costs

- Delay and customers' feedback

Techniques expected to be used:

- Descriptive statistics
- Grouped analysis
- Correlation analysis
- Distribution analysis
- Box plots
- Bar charts
- Histograms
- Scatter plots
- Heatmaps.

8.4 Project-4 Predictive Modelling

In the fourth stage, machine learning classification techniques will be used, if a delivery is likely to be delayed.

Model	Purpose
Logistic Regression	Establish an interpretable baseline
Decision Tree	Identify simple decision rules
Random Forest	Capture nonlinear relationships and interactions

Table 7. Predictive Modelling

Metrics for evaluation are as follows:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

- Furthermore, delivery_status shall not be included in the predictive features as it has a direct correspondence with the delay target variable and would create target leakage.

8.5 OPTIMIZATION STRATEGY

The process of predictive modelling can give insights into which parcels are at risk of delays. After obtaining the results of such modelling there will be made a number of recommendations for operational use.

Some of the possible areas of application may include:

1. Weather aware planning.

Enhancing reserve capacity in bad weather.

2. Route planning.

Paying special attention to long deliveries which may have a relatively high risk of delays.

3. Managing modes of delivery.

Studying the very high level of delays in respect of Express and Same Day deliveries and checking whether such delays are due to the requirements of service level or whether they are related to operational opportunities.

4. Resource allocation

Giving priority in resources to delivery segments with a higher risk of delay.

5. Communication with customers.

Using predicted risk to inform customers about delayed deliveries.

8.6 STRATEGIC DECISION FRAMEWORK



It follows that the objective of the strategy here should not be simply to maximize accuracy of prediction. The final aim is the conversion of analytics results into tangible improvements in terms of reliability of delivery, operational costs, and customer satisfaction.

9. Initial Strategic Recommendations

As per the preliminary evaluation, specific domains require more concentration in the future stage of the project. These pieces of advice have character of preliminary suggestions and should be confirmed through the following analysis and predictive modelling.

1st Suggestion — Weather Attributed Operations

Adverse weather influences the increase in the delay rate of deliveries. It is essential for the company to study weather-related contingency planning such as enhancing the number of carriers, altering delivery routes, and proactively contacting customers in regards to such deliveries.

2nd Suggestion — Study Express Delivery Performance

The percentage of delays of the express deliveries is 73.78%, which is considerably high. Instead of changing the service immediately, it is a better idea to find out whether the high percentage can be attributed to too tight delivery dates, the length of routes, oversaturation of demand in certain areas, or any other operational issue having an impact on these deliveries.

3rd Suggestion — Concentrate on Long-Distance Deliveries

The average distance of delayed deliveries equals 175.37 km against 141.30 km for on-time deliveries.

4th suggestion - Minimize the Impact of Delays on Costs

Delayed deliveries incur an average cost of \$1,016.65 while non-delayed deliveries incur a cost of \$809.75. Therefore, reducing delays can provide an opportunity to improve the operational reliability and cost efficiency of the operation.

5th suggestion - Put More Emphasis on the Customer Experience

The average rating of delayed deliveries is 2.18 compared with 4.21 for non-delayed deliveries. Thus, the reduction of delays can be considered an essential customer experience objective as well as an essential operational objective.

Recommendation	Evidence	Priority	Validation Required
Weather contingency planning	Stormy/rainy conditions show high delay rates	High	Predictive analysis
Express delivery investigation	73.78% observed delay rate	High	Deeper EDA
Long-distance risk management	175.37 km vs. 141.30 km	High	Modelling
Delay cost reduction	1,016.65 vs. 809.75	Medium–High	Cost analysis
Customer experience improvement	Rating 2.18 vs. 4.21	Medium–High	Customer analysis

Table 8. Recommendations and Validations

10. Expected Outcomes

The following work is going to contribute toward developing an organized yet insightful understanding of the delivery performance and underlying reasons for haulage problems. Additionally, the process will allow the identification of the key industry benchmarks.

The outcomes of the above-mentioned activities within this project assume:

1. Creation of a verified dataset featuring all the necessary information for analyses and procedures;
2. Performance of a thorough exploratory analysis that would support the identification of successful patterns;
3. Development of a model that would help to calculate the possibility of delay;
4. Definition of the delivery characteristics causing delay;
5. Recommendations on how to enhance the reliability and reduce the costs of the delivery process.
6. As a result, an organized data-based approach will be developed to assist managers in making the right decisions concerning route planning, resource, and delivery process management.

11. Limitations

The preliminary assessment identified a number of limitations that have to be taken into account when interpreting results.

11.1 Variables Related to Delivery Period

Both the `delivery_time_hours` and `expected_time_hours` variables in the data set have a constant value of 00:00.0, which doesn't allow for assessing the delivery duration and delays.

11.2 Repeated Delivery ID

Each delivery ID is recorded in the data base as many as 250 times. Although there are no repeated records exactly, at the point of now, it is still unclear why IDs repeat, and the data should be additionally validated during preprocessing.

11.3 Observational Relationships

As stated in the initial analysis, the associations between variables are observed, but no causal relationships can be identified. For instance, the fact that the delays increase during stormy weather doesn't mean that it is necessarily the reason for delays.

11.4 Delivery Status Leakage

Since `delivery_status` is closely linked to the delayed target, it cannot be used as a predictor in the delay classification model.

11.5 Dataset Scope

The conclusion is drawn based on the data available, and thus can't be generalized for either other logistics companies or different locations.

12. Conclusion

In this study, we provide a strategic foundation for logistics optimization driven by data, illustrated with a dataset of 25000 delivery records.

Preliminary analysis revealed a delivery delay rate of 26.68%, as well as a failure rate of 5.31%.

The following factors were identified as important for further research: weather conditions, transportation mode, and shipping distance.

Inclement weather conditions turned out to be significantly correlated with increased delay rates. Meanwhile, delayed packages tend to be exposed to larger average transportation distances and higher delivery costs.

Furthermore, delivery delays were found to significantly correlate with low customer ratings.

The analysis helped to address some essential data quality issues, such as useless fields in the delivery time data, duplicate delivery IDs, and potential target leakage using the `delivery_status` variable.

From these findings, the process will progress to include data cleaning, exploratory analysis, predictive modelling, and optimization. One of the key goals is to convert the old delivery data into useful information that can help reducing delays, improving cost efficiency, and raising the level of customers' satisfaction.

13. Reference

1. **World Bank.** *Logistics Performance Index.*
[World Bank Logistics Performance Index](#)
2. **Boysen, N., Schwerdfeger, S., & Weidinger, F.** Research on machine learning and data-driven last-mile delivery optimization.
Transportation Science.
3. Research on **data-driven optimization for last-mile delivery**, combining machine learning with routing/operational decisions.

14. Appendix: Python Code

```
import pandas as pd

import matplotlib.pyplot as plt


# =====

# PROJECT 1

# STRATEGIC PLANNING FOR DATA-DRIVEN LOGISTICS OPTIMIZATION

# =====


# =====

# 1. LOAD DATASET

# =====


df = pd.read_csv("Delivery_Logistics.csv")


print("\n===== DATASET LOADED =====")

print("Rows:", df.shape[0])

print("Columns:", df.shape[1])


# =====

# 2. DATASET OVERVIEW

# =====
```

```
print("\n===== FIRST 5 RECORDS =====")
```

```
print(df.head())
```

```
print("\n===== DATASET INFORMATION =====")
```

```
print(df.info())
```

```
print("\n===== COLUMN NAMES =====")
```

```
print(df.columns.tolist())
```

```
print("\n===== DESCRIPTIVE STATISTICS =====")
```

```
print(df.describe(include="all"))
```

```
# =====
```

```
# 3. MISSING VALUE ASSESSMENT
```

```
# =====
```

```
missing_table = pd.DataFrame({  
    "Missing_Count": df.isnull().sum(),  
    "Missing_Percentage": df.isnull().mean() * 100  
})
```

```
print("\n===== MISSING VALUE ASSESSMENT =====")
```

```
print(missing_table)
```

```
# =====
```

```
# 4. DATA TYPE ASSESSMENT
```

```
# =====
```

```
print("\n===== DATA TYPES =====")
```

```
print(df.dtypes)
```

```
# =====
```

```
# 5. UNIQUE VALUES IN TIME VARIABLES
```

```
# =====
```

```
print("\n===== DELIVERY TIME =====")
```

```
print("Data type:", df["delivery_time_hours"].dtype)
```

```
print("Unique values:", df["delivery_time_hours"].nunique())
```

```
print(df["delivery_time_hours"].value_counts(dropna=False).head())
```

```
print("\n===== EXPECTED TIME =====")
```

```
print("Data type:", df["expected_time_hours"].dtype)
```

```
print("Unique values:", df["expected_time_hours"].nunique())
```

```
print(df["expected_time_hours"].value_counts(dropna=False).head())
```

```
# =====
```

```
# 6. DELIVERY ID QUALITY CHECK
```

```
# =====
```

```
total_rows = len(df)
```

```
unique_ids = df["delivery_id"].nunique()
```

```
duplicate_id_occurrences = df["delivery_id"].duplicated().sum()
```

```

exact_duplicate_rows = df.duplicated().sum()

print("\n===== DELIVERY ID CHECK =====")
print("Total rows:", total_rows)
print("Unique delivery IDs:", unique_ids)
print("Repeated ID occurrences:", duplicate_id_occurrences)
print("Exact duplicate rows:", exact_duplicate_rows)

print("\n===== DUPLICATED DELIVERY IDs =====")

duplicate_ids = (
    df["delivery_id"]
    .value_counts()
    .loc[lambda x: x > 1]
)

print(duplicate_ids)

# =====

# 7. DELIVERY STATUS ANALYSIS

# =====

print("\n===== DELIVERY STATUS =====")
print(df["delivery_status"].value_counts())

print("\n===== DELIVERY STATUS PERCENTAGE =====")

```

```
status_percentage = (  
    df["delivery_status"]  
    .value_counts(normalize=True) * 100  
)
```

```
print(status_percentage.round(2))
```

```
# =====
```

```
# 8. CHECK RELATIONSHIP BETWEEN STATUS AND DELAY
```

```
# =====
```

```
print("\n===== DELIVERY STATUS VS DELAY =====")
```

```
status_delay = pd.crosstab(  
    df["delivery_status"],  
    df["delayed"],  
    normalize="index"  
) * 100
```

```
print(status_delay.round(2))
```

```
# =====
```

```
# 9. BASELINE KPIs
```

```
# =====
```

```
total_deliveries = len(df)
```

```
delayed_deliveries = df["delayed"].sum()
```

```
delay_rate = (  
    df["delayed"].mean() * 100  
)
```

```
failure_rate = (  
    (df["delivery_status"] == "Failed").mean() * 100  
)
```

```
average_cost = df["delivery_cost"].mean()
```

```
average_distance = df["distance_km"].mean()
```

```
average_rating = df["delivery_rating"].mean()
```

```
print("\n===== BASELINE KPIs =====")
```

```
print("Total Deliveries:", total_deliveries)
```

```
print("Delayed Deliveries:", delayed_deliveries)
```

```
print("Delay Rate:", round(delay_rate, 2), "%")
```

```
print("Failure Rate:", round(failure_rate, 2), "%")
```

```
print("Average Delivery Cost:", round(average_cost, 2))
```

```
print("Average Distance:", round(average_distance, 2), "km")
```

```
print("Average Customer Rating:", round(average_rating, 2))
```

```

# =====

# 10. KPI TABLE

# =====

kpi_table = pd.DataFrame({
    "KPI": [
        "Total Deliveries",
        "Delayed Deliveries",
        "Delay Rate (%)",
        "Failure Rate (%)",
        "Average Delivery Cost",
        "Average Distance (km)",
        "Average Customer Rating"
    ],
    "Value": [
        total_deliveries,
        delayed_deliveries,
        round(delay_rate, 2),
        round(failure_rate, 2),
        round(average_cost, 2),
        round(average_distance, 2),
        round(average_rating, 2)
    ]
})

print("\n===== KPI TABLE =====")
print(kpi_table.to_string(index=False))

```

```

# =====

# 11. DELAY RATE BY WEATHER

# =====

weather_delay = (
    df.groupby("weather_condition")["delayed"]
    .mean()
    .sort_values(ascending=False) * 100
)

print("\n===== DELAY RATE BY WEATHER =====")
print(weather_delay.round(2))

# =====

# 12. CHART — WEATHER VS DELAY

# =====

plt.figure(figsize=(9, 5))

weather_delay.plot(kind="bar")

plt.title("Delivery Delay Rate by Weather Condition")
plt.xlabel("Weather Condition")
plt.ylabel("Delay Rate (%)")
plt.xticks(rotation=0)

```

```
plt.tight_layout()
```

```
plt.savefig(  
    "delay_rate_weather.png",  
    dpi=300,  
    bbox_inches="tight"  
)
```

```
plt.show()
```

```
# =====  
# 13. DELAY RATE BY DELIVERY MODE  
# =====
```

```
mode_delay = (  
    df.groupby("delivery_mode")["delayed"]  
    .mean()  
    .sort_values(ascending=False) * 100  
)
```

```
print("\n===== DELAY RATE BY DELIVERY MODE =====")  
print(mode_delay.round(2))
```

```
# =====  
# 14. CHART — DELIVERY MODE VS DELAY
```

```
# =====

plt.figure(figsize=(9, 5))

mode_delay.plot(kind="bar")

plt.title("Delivery Delay Rate by Delivery Mode")
plt.xlabel("Delivery Mode")
plt.ylabel("Delay Rate (%)")
plt.xticks(rotation=0)

plt.tight_layout()

plt.savefig(
    "delay_rate_delivery_mode.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

# =====

# 15. AVERAGE DISTANCE BY DELAY STATUS

# =====

distance_comparison = (
    df.groupby("delayed")["distance_km"]
```

```
.mean()
)

distance_comparison.index = [
    "Not Delayed",
    "Delayed"
]

print("\n===== AVERAGE DISTANCE BY DELAY STATUS =====")
print(distance_comparison.round(2))

# =====

# 16. CHART — DISTANCE VS DELAY

# =====

plt.figure(figsize=(7, 5))

distance_comparison.plot(kind="bar")

plt.title("Average Delivery Distance by Delay Status")
plt.xlabel("Delivery Status")
plt.ylabel("Average Distance (km)")
plt.xticks(rotation=0)

plt.tight_layout()

plt.savefig(
```

```
"average_distance_delay.png",  
dpi=300,  
bbox_inches="tight"  
)
```

```
plt.show()
```

```
# =====
```

```
# 17. COST AND CUSTOMER RATING COMPARISON
```

```
# =====
```

```
comparison = df.groupby("delayed").agg({  
    "delivery_cost": "mean",  
    "distance_km": "mean",  
    "delivery_rating": "mean"  
})
```

```
comparison.index = [  
    "Not Delayed",  
    "Delayed"  
]
```

```
comparison.columns = [  
    "Average Cost",  
    "Average Distance (km)",  
    "Average Rating"  
]
```

```
print("\n===== DELAYED VS NON-DELAYED =====")
print(comparison.round(2))
```

```
# =====
# 18. VEHICLE TYPE ANALYSIS
# =====
```

```
vehicle_delay = (
    df.groupby("vehicle_type")["delayed"]
    .mean()
    .sort_values(ascending=False) * 100
)
```

```
print("\n===== DELAY RATE BY VEHICLE TYPE =====")
print(vehicle_delay.round(2))
```

```
# =====
# 19. REGION ANALYSIS
# =====
```

```
region_delay = (
    df.groupby("region")["delayed"]
    .mean()
    .sort_values(ascending=False) * 100
)
```

```
print("\n===== DELAY RATE BY REGION =====")
print(region_delay.round(2))
```

```
# =====
# 20. DELIVERY PARTNER ANALYSIS
# =====
```

```
partner_delay = (
    df.groupby("delivery_partner")["delayed"]
    .mean()
    .sort_values(ascending=False) * 100
)
```

```
print("\n===== DELAY RATE BY DELIVERY PARTNER =====")
print(partner_delay.round(2))
```

```
# =====
# 21. PACKAGE WEIGHT ANALYSIS
# =====
```

```
weight_comparison = (
    df.groupby("delayed")["package_weight_kg"]
    .mean()
)
```

```
weight_comparison.index = [
    "Not Delayed",
    "Delayed"
]

print("\n===== PACKAGE WEIGHT BY DELAY STATUS =====")
print(weight_comparison.round(2))

# =====

# 22. FINAL SUMMARY

# =====

print("\n")
print("=" * 60)
print("PROJECT 1 — FINAL ANALYTICAL SUMMARY")
print("=" * 60)

print(f"Total deliveries: {total_deliveries:,}")
print(f"Delayed deliveries: {delayed_deliveries:,}")
print(f"Delay rate: {delay_rate:.2f}%")
print(f"Failure rate: {failure_rate:.2f}%")
print(f"Average delivery cost: {average_cost:.2f}")
print(f"Average distance: {average_distance:.2f} km")
print(f"Average customer rating: {average_rating:.2f}")

print("\nProject 1 analysis completed successfully.")
```